INTRODUCTION

We have already learnt in XI standard that the feasibility of a chemical reaction under a given set of conditions can be predicted, using the principles of thermodynamics. However, thermodynamics does not provide an answer to a very important question of how fast a chemical reaction takes place. We know from our practical experience that all chemical reactions take some time for completion. Reaction speeds ranging from extremely fast (in femto seconds) to extremely slow (in years). For example, when the reactants $BaCl_2$ solution and dilute H_2SO_4 are just mixed, a white precipitate of $BaSO_4$ is immediately formed; on the other hand reactions such as rusting of Iron take many years to complete. The answers to the questions such as (i) how fast a chemical change can occur and (ii) What happens in a chemical reaction during the period between the initial stage and final stage are provided by the chemical kinetics. The word kinetics is derived from the Greek word "kinesis" meaning movement.

Chemical kinetics is the study of the rate and the mechanism of chemical reactions, proceeding under given conditions of temperature, pressure, concentration etc.

The study of chemical kinetics not only help us to determine the rate of a chemical reaction, but also useful in optimizing the process conditions of industrial manufacturing processes, organic and inorganic synthesis etc.

In this unit, we discuss the rate of a chemical reaction and the factors affecting it. We also discuss the theories of the reaction rate and temperature dependence of a chemical reaction.

7.1 Rate of a chemical reaction:

A rate is a change in a particular variable per unit time. You have already learnt in physics that change in the displacement of a particle per unit time gives its velocity. Similarly in a chemical reaction, the change in the concentration of the species involved in a chemical reaction per unit time gives the rate of a reaction.

Let us consider a simple general reaction

$$A \longrightarrow B$$

The concentration of the reactant ([A]) can be measured at different time intervals. Let the concentration of A at two different times t_2 and t_2 , $(t_2>t_1)$ be $[A_1]$ and $[A_2]$ respectively. The rate of the reaction can be expressed as

$$\text{Rate} = \frac{-[\text{Change in the concentration of the reactants}]}{(\text{Change in time})}$$

$$\text{i.e., Rate} = \frac{-([A_2]-[A_1])}{(t_2-t_1)} = -\left(\frac{\Delta[A]}{\Delta t}\right) \quad ...(7.1)$$

During the reaction, the concentration of the reactant decreases i.e. $[A_2] < [A_1]$ and hence the change in concentration $[A_2]-[A_1]$ gives a negative value. By convention the reaction rate is a positive one and hence a negative sign is introduced in the rate expression (equation 7.1)

If the reaction is followed by measuring the product concentration, the rate is given by $\left(\frac{\Delta[B]}{\Delta t}\right)$ since $[B_2]>[B_1]$, no minus sign is required here.

XII U7 kinetics - Jerald Folder.indd 205 22-12-2022 19:53:40